Lab 4: Seaweed Grazer Analysis

ALL PARTICIPATING GROUP MEMBERS HERE

Due: Wednesday, Feb 18, 8am

Part I:

1. (10 pts) Include your handwritten "cell means" two-way table describing the means for all combinations of wax and method as functions of regression parameters (βs) for the interactive model. Use an approach similar to the approach we used for the additive model to demonstrate how cells of the table are related in the the non-additive model.

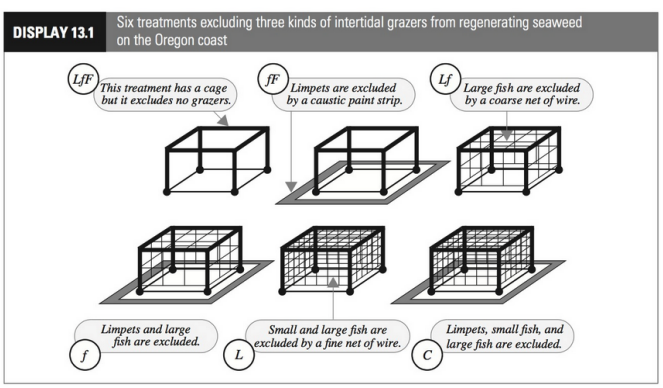
Part II: Intertidal Seaweed Grazers (Case Study)

Big-picture motivation: Interest was in investigating how the presence of 3 different ocean grazers (limpets: present (L) or not (~~L~~), small fishes: (f, ~~f~~) and large fishes: (F, ~~F~~)) affects regeneration rates of seaweed in 8 different intertidal zones (`B1:B8`). Specifically,

- Which grazer consumes the most seaweed
- Do different grazers influence impacts of the other grazers?
- Are grazing effects similar in all microhabitats (intertidal zones)?

Researchers prepared 12 plots within each intertidal zone by scraping all seaweed off of the plot and installing a randomly assigned treatment condition. Each of the 6 treatment combinations were randomly assigned to 2 sites within each block, *making this a randomized complete block design (RCBD) with 2 replications.*

- `Cover`: count (out of 100 holes) showing seaweed regeneration at the end of the study period
- `Block`: tidal condition (`B1`, `B2`,. . . ,`B8`, see text for descriptions)
- `Treat`: combination of grazers allowed to access the plot (`LfF`, `fF`, `Lf`, `f`, `L`, `C`)
- Within each `Block` there were 12 plots available. Each of the 6 treatment combinations were randomly



Block #1:	Just below high tide level, exposed to heavy surf
Block #2:	Just below high tide level, protected from the surf
Block #3:	Midtide, exposed
Block #4:	Midtide, protected
Block #5:	Just above low tide level, exposed
Block #6:	Just above low tide level, protected
Block #7:	On near-vertical rock wall, midtide level, protected
Block #8:	On near-vertical rock wall, above low tide level, protected

1. (5 pts) Explain why a non-additive model allows us to address the question of *"Are grazing effects similar in all microhabitats (intertidal zones)?"*

The code chunk below will load packages we will use and create an object named `df_grazers` that contains the data from the case study.

2. (5 pts) Fit the non-additive model and create the standard suite of diagnsotic plots. Reference appropriate plots to assess the *constant variance* assumption and *make at least one connection between the patterns you observe in the two plots you can use to make your assessment.*

When bounded counts like the variable `Count` (which are integer values bounded below by 0 and above by 100) are naively modeled as *Normal* RVs with MLR, they exhibit a particular type of non-constant variance that requires a specific transformation. *Note: Later this semester we will learn how to model counts like this with a logistic regression model (a type of generalized linear model (GLM)).* We'll use a *logit (log odds) transformation:* $logit(Cover) = log(\frac{Cover}{100 - Cover})$

3. (5 pts) Do we ever observe a count of 0 (no regeneration) or 100 (complete regeneration) in these data? What mathematical problems would arise with the *logit transformation* if we did observe either of these two extreme counts?

4. (5 pts) Create a variable for logit-regeneration within the `df_grazers` data frame.

5. (5 pts) Make a plot of logit(cover) vs. cover and describe what the logit transformation does to the counts.

6. (5 pts) Refit the non-additive model with logit-count as your new response and create the standard suite of diagnostic plots, do you have any concerns with modeling assumptions that can be assessed from these plots? Briefly explain.

7. (5 pts) Use the package `yarrr` to quickly create a raw-data plot of logit-seaweed generation as a function of block and treatment. Describe what you see.

```r
# install yarrr if needed
# install.packages(remotes)
# remotes::install_github("ndphillips/yarrr", build_vignettes = TRUE)
# load yarrr
library(yarrr)

# replace @@@@ with the variable you created for logit(Cover)
pirateplot(@@@@ ~ Block + Treat, data = df_grazers)
```

8. (5 points) Create an interaction plot using `intplot` in `catstats`. Does there appear to be an interaction between treatment and block as they relate to logit regeneration ratio? Justify your answer by describing relative features of the interaction plot.

```r
# install catstats if needed
# see above in you need to install remotes
remotes::install_github("greenwood-stat/catstats")
# use ? intplot to pull up the helpfile and see how to use the function
? intplot
```

9. (10 pts) Write code to conduct a statistical test that will allow you to address the question of *"Are grazing effects similar in all microhabitats (intertidal zones)?"* Report the output and write a summary of statistical findings to address the question. Your summary should include evidence from the statistical test and an interpretation of what that means in the context of the problem.

10. (5 pts) Generate at Type I SS ANOVA table for the additive model. Write the hypotheses associated with each row of the anova table. Which hypothesis is of interest and which would be misleading to report if this was an unbalanced design?

11. (5 pts) Make an effects plot for additive model. In what important way is this plot different from the interaction plot?

12. (5 pts) Use your plot in 10 to begin to address the question, "which grazer consumes the most seaweed?"